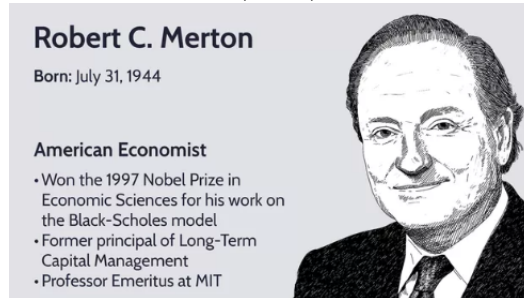


Week 10 Workshop

1. Write down an AR(2)-GARCH(1,1) model along with its underlying assumptions.

Robert Cox Merton, Nobel Laureate (1997), MIT Sloan School of Management.



2. This question is based on Merton's Intertemporal Capital Asset Pricing Model (ICAPM). The excel file *all_ord.xlsx* contains daily data on the Australian stock price index (PRICE) from 24 February 2003 to 30 September 2019. Compute the daily log returns r_t and express the log returns in percentages. Merton considers the GARCH-in-Mean model

$$\begin{aligned}r_t &= c + \gamma \sigma_t^2 + \varepsilon_t \\ \varepsilon_t &= \sigma_t u_t, \quad u_t \sim i.i.d.N(0, 1) \\ \sigma_t^2 &= \alpha_0 + \alpha_1 \varepsilon_{t-1}^2 + \beta_1 \sigma_{t-1}^2\end{aligned}$$

where γ represents the relative risk aversion parameter.

- (a) Estimate the model and write down the fitted model. In R, use the following option in the `mean.model` argument:

```
mean.model = list(  
  armaOrder = c(0, 0),  
  include.mean = TRUE,  
  archm = TRUE,  
  archpow = 2 )
```

Here, `archpow = 2` includes σ_t^2 in the mean equation, while `archpow = 1` includes σ_t .

- (b) Test the restriction $c = 0$ and interpret the result in terms of the Merton model.
 - (c) Interpret the estimate of the relative risk aversion parameter γ .
 - (d) Plot the fitted conditional variance $\hat{\sigma}_t^2$ over time and interpret its time series properties.
3. Estimate two competing asymmetric volatility models for $r_t = c + \varepsilon_t$: a GJR-GARCH model and an EGARCH model. Both models allow volatility to respond differently to positive and negative shocks. Which model fits the data better?
 4. Estimate an ARCH(1) model and an ARCH(4) model for $r_t = c + \varepsilon_t$. Which model fits the data better?